

Linux Scenario Based Questions for Interview

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1. User Management: A user complains they cannot log in. How will you troubleshoot?

Answer:

- Check if the account is locked: `passwd -S username` or `chage -l username`
- Verify account expiry: `chage -l username`
- Check `/etc/passwd` and `/etc/shadow` for account details
- Review authentication logs: `/var/log/auth.log` or `/var/log/secure`
- Test password reset: `passwd username`
- Check home directory permissions and existence
- Verify SSH configuration if remote login
- Check disk space on `/home` and `/tmp`
- Test with `su - username` from root

2. File Permissions: A script is executable by one user but not another. How do you resolve this?

Answer:

- Check current permissions: `ls -l scriptname`
- Check file ownership: `ls -l scriptname`
- Add execute permission: `chmod +x scriptname` or `chmod 755 scriptname`
- Change ownership if needed: `chown user:group scriptname`
- Check parent directory permissions
- Verify user is in correct group: `groups username`
- Check if file system is mounted with `noexec` option
- Use `sudo -u username ./script` to test execution

3. Process Management: A service is consuming 100% CPU. How will you find and fix it?

Answer:

- Identify the process: `top`, `htop`, or `ps aux --sort=-%cpu`
- Get detailed process info: `ps -ef | grep processname`
- Check process tree: `pstree -p PID`
- Analyze with: `strace -p PID` (system calls)
- Lower priority: `nice -n 19 command` or `renice 19 PID`

- Kill if necessary: kill PID or kill -9 PID
- Restart the service: systemctl restart servicename
- Check service logs: journalctl -u servicename
- Monitor resource usage: iotop, iftop

4. SSH Issues: You cannot SSH into a remote machine. How do you debug?

Answer:

- Test basic connectivity: ping hostname
- Check if SSH port is open: telnet hostname 22 or nmap -p 22 hostname
- Try verbose SSH: ssh -v username@hostname
- Check SSH client config: ~/.ssh/config
- Verify SSH keys: ssh-keygen -l -f ~/.ssh/id_rsa.pub
- Check server SSH logs: /var/log/auth.log
- Verify SSH daemon is running: systemctl status sshd
- Check SSH configuration: /etc/ssh/sshd_config
- Test with password if key fails: ssh -o PreferredAuthentications=password

5. Disk Space Full: / partition is full. How do you find and delete large files safely?

Answer:

- Check disk usage: df -h
- Find largest directories: du -sh /* | sort -hr
- Find large files: find / -type f -size +100M -exec ls -lh {} \;
- Clean temporary files: rm -rf /tmp/* and /var/tmp/*
- Clean logs: journalctl --vacuum-time=7d
- Clear package cache: yum clean all or apt clean
- Check for deleted but open files: lsof +L1
- Remove old kernels: package-cleanup --oldkernels --count=2
- Clean user trash and browser cache

6. File Corruption: A log file is showing junk characters. How will you check and recover it?

Answer:

- Check file encoding: file filename
- Try different character sets: iconv -f ISO-8859-1 -t UTF-8 filename
- Check for binary data: hexdump -C filename | head

- Verify file integrity: `md5sum filename` (if original checksum available)
- Check file system: `fsck /dev/device`
- Restore from backup if available
- Check if file is actively being written: `lsof filename`
- Try to salvage readable parts: `strings filename > recovered.log`
- Stop the service writing to it: `systemctl stop servicename`

7. Crontab Not Running: A scheduled job is not executing. How do you debug?

Answer:

- Check if cron daemon is running: `systemctl status crond`
- View user's crontab: `crontab -l`
- Check cron logs: `/var/log/cron` or `journalctl -u cron`
- Verify cron syntax: Use online cron validators
- Test the command manually in shell
- Check script permissions and paths
- Ensure full paths in cron commands
- Check environment variables: `env` in cron vs shell
- Verify user has cron access: `/etc/cron.allow` and `/etc/cron.deny`

8. Package Installation Failing: yum or apt is failing. How will you resolve it?

Answer:

- Check repository configuration: `/etc/yum.repos.d/` or `/etc/apt/sources.list`
- Update package database: `yum update` or `apt update`
- Check network connectivity to repositories
- Clear package cache: `yum clean all` or `apt clean`
- Check disk space: `df -h`
- Resolve dependency conflicts: `yum deplist package` or `apt-cache depends`
- Try force reinstall: `yum reinstall` or `apt --reinstall install`
- Check for held packages: `apt-mark showhold`
- Use alternative package managers: `dnf` instead of `yum`

9. User Cannot Sudo: A user was added to sudoers, but sudo still doesn't work. What could be wrong?

Answer:

- Check sudoers file syntax: `visudo -c`
- Verify user is in sudo group: `groups username`
- Check sudoers file: `cat /etc/sudoers`
- Look for user-specific rules: `/etc/sudoers.d/`
- Test sudo access: `sudo -l -U username`
- Check for typos in username or permissions
- Verify user session: logout and login again
- Check if sudoers file is corrupted
- Use `pkexec visudo` if sudo is completely broken

10. Kernel Panic: How do you troubleshoot and recover from a kernel panic?

Answer:

- Boot from rescue/recovery media
- Check kernel panic message from console or `/var/log/messages`
- Boot with previous working kernel from GRUB menu
- Check hardware: memory test, disk check
- Verify recent kernel updates: `rpm -qa kernel` or `dpkg -l linux-image*`
- Check for hardware compatibility issues
- Boot in single-user mode: add `single` to kernel parameters
- Rollback recent changes or updates
- Check disk space and file system integrity
- Update initramfs: `dracut -f` or `update-initramfs -u`

11. Zombie Processes: How do you identify and remove zombie processes?

Answer:

- Identify zombies: `ps aux | grep '<defunct>'` or `ps -eo stat,pid,ppid,comm | grep '^Z'`
- Check parent process: `ps -ef | grep PPID`
- Kill parent process: kill PPID (zombies will be inherited by init)
- Use signal to parent: `kill -CHLD PPID`
- Restart parent service if possible
- Check system load: `uptime`
- Monitor with: `top` (look for 'Z' state)
- Prevent by fixing parent process to properly handle `SIGCHLD`

- Reboot as last resort if system becomes unstable

12. Too Many Open Files Error: How do you fix "Too many open files" in Linux?

Answer:

- Check current limits: `ulimit -n` and `ulimit -Hn`
- Check system-wide limits: `/proc/sys/fs/file-max`
- Increase user limits in `/etc/security/limits.conf`:

text

`username soft nfile 4096`

`username hard nfile 8192`

- Check current open files: `lsuf | wc -l`
- Find processes with most open files: `lsuf | awk '{print $2}' | sort | uniq -c | sort -nr`
- Restart services after changing limits
- Check application configuration for connection pools
- Modify systemd service limits if needed

13. Finding and Killing Processes: A process is causing high memory usage. How do you locate and stop it?

Answer:

- Find memory-heavy processes: `ps aux --sort=-%mem | head`
- Use `top`/`htop`: `top` then press 'M' to sort by memory
- Get detailed memory info: `cat /proc/PID/status`
- Check memory maps: `pmap PID`
- Kill gracefully: `kill PID`
- Force kill if needed: `kill -9 PID`
- Kill by name: `pkill processname`
- Kill all user processes: `pkill -u username`
- Monitor system memory: `free -h` and `vmstat`

14. Home Directory Missing: A user's home directory is missing. How will you restore it?

Answer:

- Check if user exists: `id username`
- Create home directory: `mkdir /home/username`
- Copy skeleton files: `cp -r /etc/skel/. /home/username/`

- Set ownership: `chown -R username:username /home/username`
- Set permissions: `chmod 755 /home/username`
- Update user's home in `passwd`: `usermod -d /home/username username`
- Restore from backup if available
- Check if home is on different partition: `mount | grep home`
- Verify user can login and access files

15. Time Sync Issues: The server time is incorrect. How do you fix NTP sync?

Answer:

- Check current time: `date` and `timedatectl`
- Check NTP service: `systemctl status ntp` or `systemctl status chronyd`
- Check NTP servers: `ntpq -p` or `chrony sources`
- Manual time sync: `ntpdate pool.ntp.org`
- Configure NTP: `/etc/ntp.conf` or `/etc/chrony.conf`
- Set timezone: `timedatectl set-timezone America/New_York`
- Enable NTP: `timedatectl set-ntp true`
- Restart NTP service: `systemctl restart ntp`
- Check hardware clock: `hwclock --show`

16. Finding Recently Changed Files: How do you find all files modified in the last 10 minutes?

Answer:

- Use find command: `find /path -type f -mmin -10`
- For specific directory: `find /var/log -mmin -10`
- Include access time: `find /path -type f -amin -10`
- Show detailed info: `find /path -type f -mmin -10 -ls`
- Exclude certain directories: `find /path -type f -mmin -10 -not -path "*/proc/*"`
- Sort by modification time: `find /path -type f -mmin -10 -exec ls -lt {} +`
- Monitor real-time changes: `inotifywatch /path`

17. Deleting Large Files But Space Not Freeing Up: What could be the reason?

Answer:

- Check for open file handles: `lsof +L1` (deleted but still open files)
- Find processes holding deleted files: `lsof | grep deleted`
- Restart services holding the files: `systemctl restart servicename`

- Check if files are hard-linked: `ls -li filename`
- Verify you deleted from correct partition: `df -h`
- Check for hidden files: `ls -la`
- Empty trash/recycle bin
- Check if deletion was successful: `ls -l filename`
- Kill processes holding the file handles: `kill PID`

18. Checking System Load: How do you analyze high system load issues?

Answer:

- Check load average: `uptime` and `w`
- Monitor with `top`/`htop`: `top` or `htop`
- Check I/O wait: `iostat -x 1`
- Monitor disk activity: `iostat`
- Check memory usage: `free -h` and `vmstat`
- Network monitoring: `iftop` or `nethogs`
- Check for CPU-bound processes: `ps aux --sort=-%cpu`
- System activity: `sar -u 1 10`
- Check for swap usage: `swapon -s`

19. NFS Mount Issues: How do you troubleshoot an NFS mount not working?

Answer:

- Check NFS services: `systemctl status nfs-server rpcbind`
- Test NFS server connectivity: `showmount -e nfs-server`
- Check exports: `cat /etc/exports`
- Verify mount command: `mount -t nfs server:/path /mnt/point`
- Check network connectivity: `ping nfs-server`
- Test RPC services: `rpcinfo -p nfs-server`
- Check firewall rules: `iptables -L` or `firewall-cmd --list-all`
- Check NFS logs: `/var/log/messages`
- Verify permissions on export directory
- Try different NFS versions: `mount -t nfs -o vers=3`

20. Hostname Resolution Issues: ping is working, but ssh is failing by hostname. Why?

Answer:

- Check DNS resolution: nslookup hostname or dig hostname
- Check /etc/hosts file for hostname entries
- Check /etc/nsswitch.conf for name resolution order
- Test reverse DNS: dig -x IP_ADDRESS
- Check SSH configuration: /etc/ssh/sshd_config (UseDNS setting)
- Try SSH with IP instead: ssh user@IP_ADDRESS
- Check if IPv6 is causing issues: ssh -4 user@hostname
- Verify hostname matches certificate if using SSH keys
- Check SSH client configuration: ~/.ssh/config

21. How do you extend a partition without unmounting it?

Answer:

- For LVM volumes:

bash

lvextend -L +10G /dev/vg/lv

resize2fs /dev/vg/lv # for ext2/3/4

xfs_growfs /mount/point # for XFS

- Check current size: df -h and lsblk
- Extend physical volume first if needed: pvextend /dev/sda1
- For non-LVM: Use growpart utility
- Verify file system type: mount | grep partition
- Online resize for ext4: resize2fs /dev/device
- For XFS: xfs_growfs /mount/point
- Check after extension: df -h

22. What steps are needed to add a new disk to a Linux server?

Answer:

- Physical connection and detection: lsblk or fdisk -l
- Partition the disk: fdisk /dev/sdX or parted /dev/sdX
- Create file system: mkfs.ext4 /dev/sdX1
- Create mount point: mkdir /mnt/newdisk
- Mount temporarily: mount /dev/sdX1 /mnt/newdisk
- Add to fstab: echo "/dev/sdX1 /mnt/newdisk ext4 defaults 0 2" >> /etc/fstab

- Test fstab: `mount -a`
- Set permissions: `chmod 755 /mnt/newdisk`
- Verify: `df -h` and `lsblk`

23. How to check which processes are writing to a file in real time?

Answer:

- Use lsof: `lsof filename`
- Monitor file access: `lsof +f -- filename`
- Use inotify: `inotifywait -m filename`
- Check with fuser: `fuser -v filename`
- Monitor directory: `inotifywait -m -r /path/to/directory`
- Use strace on process: `strace -e write -p PID`
- System-wide monitoring: `iotop -o`
- Check file locks: `lsofs`
- Monitor with auditd: `auditctl -w /path/to/file -p wa`

24. How to recover a deleted file in Linux?

Answer:

- Stop writing to the disk immediately
- Check if file is in trash: `~/.local/share/Trash/`
- Use file recovery tools:
 - `testdisk` and `photorec`
 - `extundelete` for ext3/4
 - `debugfs` for ext2/3/4
- Check for recent backups
- Look for temporary copies: `/tmp`, `/var/tmp`
- Check application-specific recovery:
 - Database logs
 - Editor backup files
- Use `grep` to search for file content in raw disk
- Restore from system snapshots if available

25. How to check disk I/O performance in Linux?

Answer:

- Use iostat: `iostat -x 1 5`
- Monitor with iotop: `iotop -o`
- Check with sar: `sar -d 1 10`
- Use dd for testing: `dd if=/dev/zero of=testfile bs=1M count=1000`
- hdparm for disk tests: `hdparm -t /dev/sdX`
- Use fio for comprehensive testing:

bash

`fio --name=randwrite --ioengine=libaio --bs=4k --rw=randwrite --size=1G`

- Check disk usage: `df -h` and `du -sh`
- Monitor system load: `uptime`
- Use atop for detailed monitoring

26. A directory is taking too much space. How do you analyze it?

Answer:

- Check directory size: `du -sh /path/to/directory`
- Find largest subdirectories: `du -sh /path/* | sort -hr`
- Recursive analysis: `du -ah /path | sort -hr | head -20`
- Use ncdu for interactive analysis: `ncdu /path`
- Find large files: `find /path -type f -size +100M -exec ls -lh {} \;`
- Check hidden files: `du -ah /path/.[^.]* | sort -hr`
- Use tree command: `tree -h /path`
- Find old files: `find /path -type f -mtime +30 -ls`
- Check for duplicate files: `fdupes -r /path`

27. How do you remount a filesystem in read-write mode without rebooting?

Answer:

- Remount as read-write: `mount -o remount,rw /mountpoint`
- Check current mount options: `mount | grep mountpoint`
- For root filesystem: `mount -o remount,rw /`
- Check file system errors first: `fsck /dev/device` (if safe)
- Verify no processes are using it: `lsof +f -- /mountpoint`
- Check fstab entry: `grep mountpoint /etc/fstab`
- Force remount if needed: `mount -o remount,rw,force /mountpoint`

- Verify successful remount: `mount | grep mountpoint`
- Check write permissions: `touch /mountpoint/testfile`

28. What happens when a file is deleted but is still in use by a process?

Answer:

- File descriptor remains open in process
- Disk space is not freed until process closes file or terminates
- File becomes "deleted" but still accessible to the process
- Check with: `ls -l` (shows deleted files still open)
- File appears as "(deleted)" in `ls -l` output
- Process can continue reading/writing to the file
- To free space: kill the process or restart it
- Check which process: `ls -l | grep deleted`
- File will be completely removed when last reference is closed
- This is common with log files and temporary files

29. How do you create and mount a swap file?

Answer:

- Create swap file: `dd if=/dev/zero of=/swapfile bs=1M count=2048`
- Set permissions: `chmod 600 /swapfile`
- Format as swap: `mkswap /swapfile`
- Enable swap: `swapon /swapfile`
- Add to fstab: `echo "/swapfile none swap sw 0 0" >> /etc/fstab`
- Verify swap is active: `swapon -s` or `free -h`
- Check swap usage: `cat /proc/swaps`
- Set swappiness: `echo 'vm.swappiness=10' >> /etc/sysctl.conf`
- Apply sysctl changes: `sysctl -p`

30. How do you find large unused files across multiple partitions?

Answer:

- Find large files: `find / -type f -size +100M -atime +30 2>/dev/null`
- Check all partitions: `df -h` then search each mount point
- Find by size and access time: `find / -type f -size +1G -atime +60 -ls`
- Exclude system directories:

bash

find / -type f -size +100M -atime +30 -not -path "/proc/*" -not -path "/sys/*"

- Use locate database: locate -S then locate -r '.*' | xargs ls -lah | grep '^.*[0-9]G'
- Find temporary files: find /tmp /var/tmp -type f -size +100M -atime +7
- Check log files: find /var/log -type f -size +100M -mtime +30

31. How do you fix a corrupted file system using fsck?

Answer:

- Unmount the filesystem first: umount /dev/sdX1
- Run fsck in check-only mode: fsck -n /dev/sdX1
- Run interactive repair: fsck /dev/sdX1
- Force check: fsck -f /dev/sdX1
- Auto-repair: fsck -y /dev/sdX1 (answer yes to all)
- For specific file systems:
 - ext2/3/4: e2fsck -f /dev/sdX1
 - XFS: xfs_repair /dev/sdX1
- Check bad blocks: fsck -c /dev/sdX1
- Boot from rescue media for root filesystem
- Always backup important data first if possible

32. What is inode exhaustion, and how do you resolve it?

Answer:

- Check inode usage: df -i
- Find directories with many files: find / -xdev -type f | cut -d/ -f2 | sort | uniq -c | sort -n
- Find directories with most files: find /path -type d -exec sh -c 'echo "\$(ls -1A "\$1" | wc -l) \$1"' _ {} \; | sort -n
- Delete unnecessary files, especially:
 - Temporary files in /tmp
 - Old log files
 - Cache files
- Clean package manager cache
- Remove old kernels and headers
- Check for directories with many small files

- Increase inode count (requires filesystem recreation):

bash

```
mkfs.ext4 -N 2000000 /dev/sdX1
```

33. You need to copy a huge file across servers. What's the fastest way?

Answer:

- Use rsync with compression: `rsync -avz --progress file user@server:/path/`
- Use scp with compression: `scp -C largefile user@server:/path/`
- Use netcat for direct transfer:
 - Receiver: `nc -l 9999 > file`
 - Sender: `nc server 9999 < file`
- Use tar with ssh: `tar -czf - file | ssh user@server 'tar -xzf - -C /path/'`
- Parallel transfer with aria2: `aria2c --file-allocation=none file`
- Use bbcp for high-speed transfers: `bbcp file user@server:/path/`
- Consider splitting large files: `split -b 1G file prefix`
- Use compression if network is slow: gzip file before transfer

34. Your `df -h` and `du -sh` show different disk usage. Why?

Answer:

- Deleted files still held open by processes (check with `lsof +L1`)
- Hard links counted multiple times by `du`
- Sparse files (`du` shows actual disk usage, `df` shows allocated space)
- Different block size calculations
- Files in mount points hidden by mounted filesystems
- Metadata overhead (inodes, journal) counted by `df`
- Snapshot or backup files taking space
- Check for files in directories that are mount points
- Reserved space for root user (usually 5% on ext filesystems)
- Different units or rounding differences

35. How do you find out which directory is consuming the most space in `/var`?

Answer:

- Use `du` command: `du -sh /var/* | sort -hr`
- Drill down into largest directory: `du -sh /var/log/* | sort -hr`

- Interactive exploration: `ncdu /var`
- Show all subdirectories: `du -ah /var | sort -hr | head -20`
- Find large files specifically: `find /var -type f -size +100M -exec ls -lh {} \;`
- Use tree with sizes: `tree -h -L 2 /var`
- Check specific common directories:

bash

`du -sh /var/log /var/cache /var/lib /var/spool /var/tmp`

- Real-time monitoring: `watch "du -sh /var/* | sort -hr"`